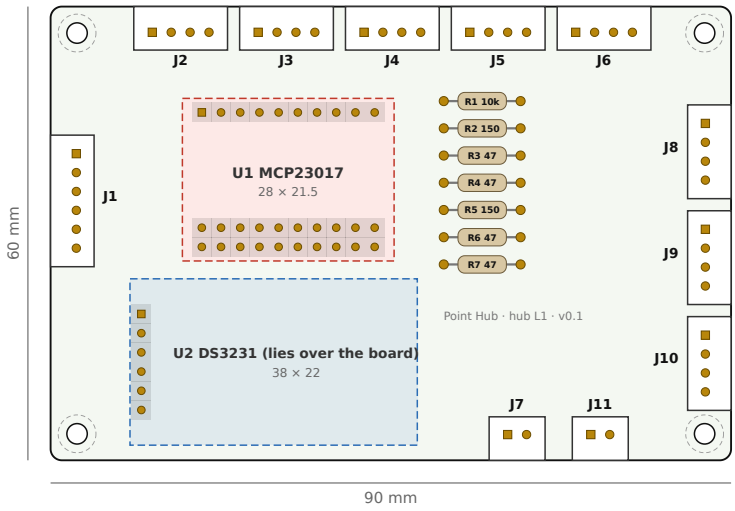
Hub board, Level 1 — paper check at 1:1

Point Hub board · v0.1 · 12.11.2026 · seen from the parts side · print at 100 %



50 mm: check with the ruler before you lay parts on it

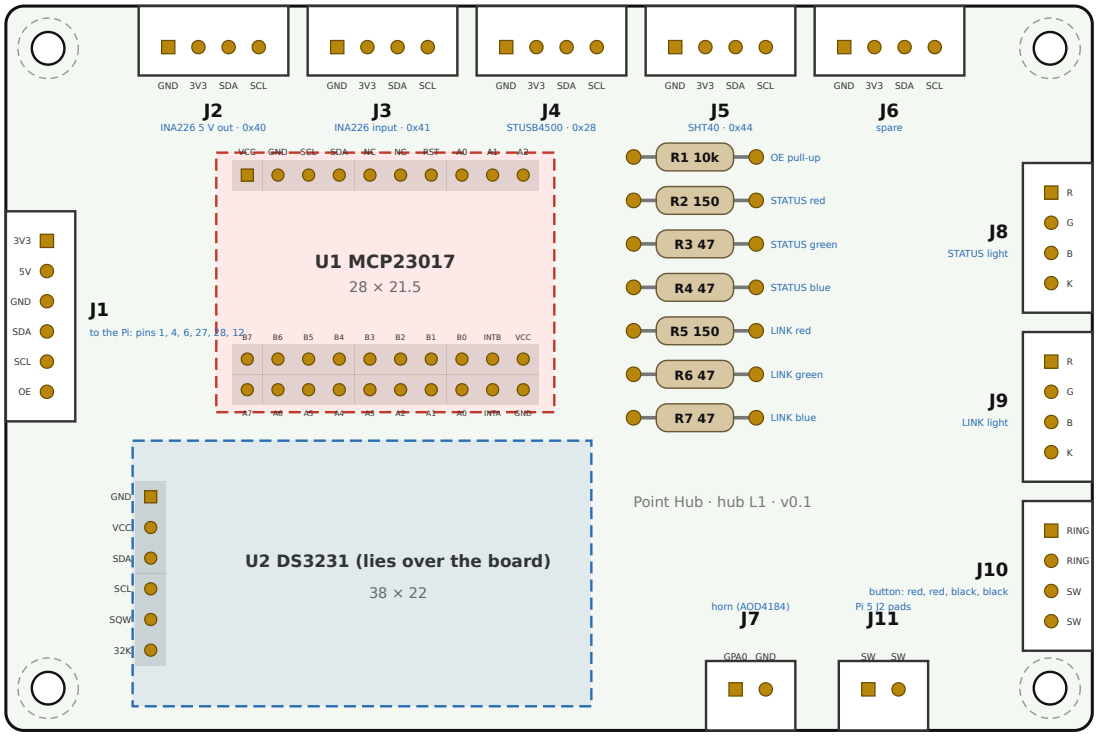
What to check

1. Print page 1 at **100 % / actual size** (not “fit to page”). The 50 mm bar must measure 50 mm, the board 90 × 60.
2. Lay the **MCP23017** on U1 with its pins over the squares: does the single row fall on the top row, and is **VCC** at the left end? Which double row is **A** (with GND at the right end): the inner or the outer one?
3. Lay the **DS3231** on U2: header on the six squares, board to the right. Is **GND** the top pin? Does the cell holder face down or up?
4. Put the **INA226**, **AOD4184** and **button** next to the paper where they will sit in the box: are the J-sockets on the right side for their wires?
5. Tell me what moves; I make the KiCad board from the corrected positions.

Squares = pin 1. Red outline = MCP23017 (plugs in from above). Blue outline = DS3231 (on its right-angle header, lying over the board). Dashed circles = M2.5 holes, keep clear. The INA226 boards, AOD4184, PD board and button are **not** on the hub; they connect with cables.

Pin names (enlarged, not for measuring)

Socket pin order as seen from the parts side; pin 1 is the square pad.



Notes

1. Everything runs at **3.3 V** (MCP23017, DS3231, INA226, SHT40). The only 5 V on the hub is the button ring.
2. I²C pull-ups come from the modules. With all of them on the bus they add up: if the bus is unreliable, cut the I²C pull-up jumper on the STUSB4500 board first.
3. The I²C ports are JST-XH, not Qwiic: through-hole, easy to solder by hand, and they lock. Wire each pigtail to the module's own pin labels (the INA226 order is VCC SDA SCL ALE GND).
4. The J1 lead: XH-6 on the hub, six single Dupont housings on the Pi's pins 1, 4, 6, 12, 27, 28 through the tall stacking header.
5. Positions follow ruler photos (±1 mm). Nothing is ordered until the paper check is done.

Connections

Net	Joins
3V3	J1·1 (Pi pin 1) → every port J2–J6 pin 2, U1 VCC (both), U1 RST, U2 VCC, R1
5V	J1·2 (Pi pin 4) → J10·1 (button ring)
GND	J1·3 (Pi pin 6) → ports J2–J6 pin 1, U1 GND (both), U1 address pins A0 A1 A2 (single row), U2 GND, J7·2, J8·4, J9·4, J10·2
SDA	J1·4 (Pi pin 27, GPIO0) → ports J2–J6 pin 3, U1 SDA, U2 SDA
SCL	J1·5 (Pi pin 28, GPIO1) → ports J2–J6 pin 4, U1 SCL, U2 SCL
OE	J1·6 (Pi pin 12, GPIO18) → R1 10k → 3V3: panels stay dark at power-on
Horn	U1 GPA0 (port-A row “A0”) → J7·1 (horn: AOD4184 PWM; J7·2 → its GND)
STATUS	U1 GPA1 → R2 150 → J8·1 red; GPA2 → R3 47 → J8·2 green; GPA3 → R4 47 → J8·3 blue; J8·4 = common cathode
LINK	U1 GPA4 → R5 150 → J9·1 red; GPA5 → R6 47 → J9·2 green; GPA6 → R7 47 → J9·3 blue; J9·4 = common cathode
Button	J10·3 → J11·1, J10·4 → J11·2 (straight through to the Pi 5 J2 pads; either way round)
Unused	U1 port B, INTA, INTB, GPA7 (input); U2 SQW, 32K

Ports

Socket	Pins	Goes to
J1	3V3 · 5V · GND · SDA · SCL · OE	to the Pi: pins 1, 4, 6, 27, 28, 12
J2	GND · 3V3 · SDA · SCL	INA226 5 V out · 0x40
J3	GND · 3V3 · SDA · SCL	INA226 input · 0x41
J4	GND · 3V3 · SDA · SCL	STUSB4500 · 0x28
J5	GND · 3V3 · SDA · SCL	SHT40 · 0x44
J6	GND · 3V3 · SDA · SCL	spare
J7	GPA0 · GND	horn (AOD4184)
J8	R · G · B · K	STATUS light
J9	R · G · B · K	LINK light
J10	RING · RING · SW · SW	button: red, red, black, black
J11	SW · SW	Pi 5 J2 pads

Parts for the hub

Part	How many	State
Hub PCB	90 × 60, 2 layers, 1.6 mm, JLCPCB (5 pieces is the minimum order)	after this check, in KiCad
JST-XH sockets, top entry, through-hole	2-pin × 2, 4-pin × 8, 6-pin × 1	to buy
JST-XH pigtails, pre-crimped, 20–30 cm	2-pin × 2, 4-pin × 8, 6-pin × 1 (no crimping needed)	to buy
Female headers 2.54 mm	1 × 10 (×1), 2 × 10 (×1), 1 × 6 (×1)	to buy (or from a kit)
Dupont female housings 1-pin	6, for the Pi end of the J1 lead	to buy (or cut a jumper)
Resistors	10k × 1, 150 × 2, 47 × 4	arrived 12.11
M2.5 standoffs + screws	4	kit